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TDRLM: Stylometric learning for authorship verification by Topic-Debiasing

Xinyu Hu , Weihan Ou , Sudipta Acharya , Steven H.H. Ding , Ryan D'Gama , Hanbo Yu

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Abstract

Authorship verification is the task of determining if a candidate is the real author of the provided sequences of texts. Stylometric learning is the study of the writing style by analyzing the textual data through computational approaches for many linguistic tasks such as authorship verification, authorship identification, etc. In recent years, Deep learning has been used as an effective solution to stylometric learning for authorship verification. However, most of the existing studies do not consider the semantic biases captured by the model, and the learned stylometric features include a strong indicator of the topical biases. During the extraction of stylometric features, we found that specific words related to the latent topics impact the performance of machine learning algorithms and, therefore, are not generalizable toward out-of-sample authors. In this paper, we propose a latent topic score dictionary with a Topic-Debiasing Representation Learning Model (TDRLM) for stylometric representation learning on the problem of authorship verification. The model applies position-specific topic scores on a topic-debiasing attention mechanism in order to adjust the tokenized texts based on the topical bias. The experimental results show that our proposed approach achieves the best performance with the highest Area Under Curve (AUC) of 92.47% for the *Twitter-Foursquare* dataset and 93.11% for the (ICWSM) Twitter dataset, both better than the state-of-the-art stylometric learning models and the up-trained language models.

Introduction

Authorship verification (Koppel & Schler, 2004) is the problem of verifying whether a candidate is the real author of the provided texts or not by analyzing the stylistic traits in the textual data. With the prevalence of excessive Internet usage, social media has become one of the main news sources (Newman, Levy, & Nielsen, 2015), and studies have shown that false information spread faster than the truth stories.¹ Through account hacking or identity impersonation, malicious actors have started leveraging such an opportunity to spread falsified or sensitive information for the purpose of financial gain, political manipulation, brand image manipulation, and personal attacks. Authorship verification has the potential to unveil the attackers behind the misinformation and trace the attacks to their origins by inspecting the stylistic traits found in the text data and verifying them against a known candidate profile.

Authorship verification has a long-standing history in computational linguistics research (Iqbal, Khan, Fung, & Debbabi, 2010). Before the computational era, linguistic experts manually conducted the authorship verification task by inspecting stylistic traits according to their domain knowledge (Boenninghoff, Hessler, Kolossa, & Nickel, 2019). As the development of data-driven solutions became prevalent across sectors, supervised learning-based computational models such as Support Vector Machine (SVM) have been applied to address the problem of authorship verification with a higher degree of accuracy and efficiency (Manevitz & Yousef, 2001). Later, deep learning and representation learning models, such as Bidirectional Encoder Representations from Transformers (Devlin, Chang, Lee, & Toutanova, 2018), and Sentence BERT (Reimers & Gurevych, 2019) show promising results for semantic learning in Natural Language Processing (NLP). Following similar ideas, many studies have been conducted for stylometric feature learning (Ding et al., 2017, Halvani et al., 2020, Rivera-Soto et al., 2021). Ding et al. proposed learning models that include various types of linguistic aspects in distributed representations of words to simultaneously learn the representations of writing style from unlabeled texts for authorship analysis. Recently, Halvani et al. presented an alternative authorship verification method that bases its classification decision only on topic-agnostic features such as “wanna”, “gotta” which can be seen as part of someone’s personal preferred writing style, instead of domain-specific words such as “i7”. Also, Rivera-Soto et al. carried out the first comprehensive research of cross-domain transfer learning for authorship verification, taking into account zero-shot transfers involving three different domains.

These studies have been able to demonstrate the potential of deep learning solutions for authorship verification. All of them have achieved promising results in different settings. However, we still noted certain limitations. We observe that the existing stylometric representations may be biased toward the specific topical preference. Although Ding et al., Halvani et al., and Rivera-Soto et al. also noticed this issue, their proposed models still provide a limited capability in handling topical bias.

Ding et al. separated topical and lexical preferences through gradient flows. However, the topical information can still be present in the lexical preference representation. The topic-related words are used within the input, and there is no guarantee that the gradient toward lexical preference will be zero.

Moreover, even though Rivera-Soto et al. removed the bias of the domain defined by the specific social media platforms such as Reddit and Amazon, the matching between users within the same domain can still be subject to topical bias. On the other hand, Halvani et al. proposed an extreme case solution that covers only stop-word n-gram features under different syntactic categories. However, it will miss non-standard English stylistic words, such as 'OMG' or 'LOL', that will contribute to topic-agnostic features. Based on this observation, we designed a new approach to remove topical bias in order to provide better out-of-sample author generalizability.

We consider the textual data in the form of natural language as a mixture of personal preferences on word choice and prescribed topics that may be affected by external factors or contextual intentions. Topical preference could be a good writing style indicator, but most of the time it would incur false positives and false negatives when there is a drift in the topic discussed. Especially on social media where a high diversity of topics can be observed and topical drift happens along with evolving social and political events. For example, in one post a Twitter user may discuss computer products such as "*I wanna buy a Alienware x14 gaming laptop produced by Dell because it looks so beautiful and is very well configured with the processor of 12th Gen Intel i7*". And later we will need to match to another post that talks about a different topic such as "*2022 Present Election*". The stylometric learning method should be able to avoid picking up the topical biases and avoid semantics-related word sequences such as '12th Gen Intel i7'. It should be noted that the word 'Gen' itself may or may not carry topical bias. It is the context of the post that ultimately determines the actual meaning and function of the word.

Motivated by the above observation, we propose a topic score dictionary to measure the context-dependent topic impact score for each word leveraging sub-word tokenization and Latent Dirichlet Allocation (LDA). We estimate how likely a word carries topical bias by looking through the prior probability that it can be associated with a specific topic combined with the semantic roles it plays in the current context. Afterward, we present a Topic-Debiasing Representation Learning Model (TDRML) by combining a multi-head attention mechanism, the topic score dictionary, and a pre-trained language model. By removing topic-related bias through the attention mechanism, the learned stylometric representation is capable of matching tweets that are of dramatic difference in the topics discussed.

For the experimental study, we benchmark 15 state-of-the-art learning-based methods such as different types of n-gram (Guthrie, Allison, Liu, Guthrie, & Wilks, 2006), word2vec (Mikolov, Chen, Corrado, & Dean, 2013), LDA (Blei, Ng, & Jordan, 2003), and all-distilroberta-v1 (Reimers, 2019), etc. We use two social media datasets: *ICWSM* and *Twitter-Foursquare*. They are both social media postings that cover a high degree of stylistic and topical variance. To evaluate the effectiveness of models in different scenarios, we create three different setups of sample combinations to control the amount of available information in the verification stage such as one-sample, two-sample, and three-sample combinations. The experimental results indicate that TDRML achieved the best model performance with both validation set and testing set experiments for the six scenarios of the *ICWSM* and *Twitter-Foursquare* dataset. Overall, three major contributions of this research are described as follows.

1. We propose a topic score dictionary to record the prior probability of a sub-word token observed in a post being a result of the context-specific topical choice.
2. By embedding the topic score dictionary into an attention mechanism, we develop the Topic-Debiasing Representation Learning Model (TDRLM) to remove context-specific topical bias from the learned stylometric representation.
3. We benchmark state-of-the-art learning-based methods for authorship verification using two social media datasets with three different sample combination setups to access their performance when exposed to a limited amount of information. The result indicates that such a topic-debiasing design has a significant impact on model performance, and our proposed model TDRLM achieves the highest *AUC* score of 92.56% compared to the other 15 methods.

The rest of the paper is organized as follows. Section 2 describes existing relevant works for authorship verification. Section 3 defines and formally formulates the problem under study. Section 4 explains and justifies the detailed design of our proposed method. Next, Section 5 describes the experimental setups and details of the datasets. A detailed analysis of the obtained results is provided in Section 6. Finally, Section 7 concludes the paper and discusses future directions.

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Section snippets

Related works

Authorship Verification: Many features such as voice features, and semantic features, have been gradually developed in recent years to improve human living quality. For example, Siri responds to user requests based on voice features (Burbach et al., 2019). Except this, stylometric features are observed as a type of critical linguistic features for authorship verification because it is able to represent personal characteristics in linguistics (Brennan et al., 2012, Stamatatos, 2009). Unlike ...

Problem definition

The authorship verification problem can be formulated as follows. Given two lists of tweets from a known dataset, denoted as $\mathbf{t}^1$ and $\mathbf{t}^2$, our task is to compare them and determine whether they are written by the same author or not. Here, $\mathbf{t}^1 = (t_1^1, t_2^1, \dots, t_n^1)$ denotes a sequence of tweets all

authored by a given user, and $\mathbf{t}^2 = (t_1^2, t_2^2, \dots, t_n^2)$ denotes another sequence of tweets from the same user or a different user than $\mathbf{t}^1$. When n equals 1, we have the most challenging 1-combination scenario where we ...

Methodology

In this section, we explain the inner working methodologies of the proposed model for stylometric representation learning. We will begin with a broad overview of the architecture, followed by a detailed illustration of the three bottom-up components: Topic Modeling, Topic-Debiasing Representation Learning Model (TDRLM), and Similarity Learning. Topic Modeling pre-processes and creates Topic Score Dictionary from the training set. TDRLM removes the topical bias from tweets by attention. At last, ...

Experiment

This section describes the overall experiment by introducing the baseline methods, datasets, and experimental settings. To evaluate our model through comparative analysis, we prepared 6 basic baselines and 5 complex baselines (named as baseline models) in total. The first 6 basic baselines are described in brief as follows.

- **{1 to 5}-n-grams:** The first baseline uses n-grams (Guthrie et al., 2006). It is a way of word embedding creation and can be called n-word-grams. It represents a sequence of N ...

...

Results and discussion

This section presents the experimental results and analysis of baselines and our proposed model TDRLM. Six tables (Table 1–Table 6) display the overall results of our experiments with respect to four evaluation metrics for both of *ICWSM* and *Twitter-Foursquare* datasets under three different types of sample combinations: One tweet per sample, Two tweets per sample, and Three tweets per sample. To analyze the results well, *AUC* is seen as the most important one among all the evaluation metrics. ...

Conclusion

In a nutshell, we proposed the Topic Score Dictionary and TDRLM to achieve the rational topical scale used on the authorship verification based on stylometric learning. Topic Score Dictionary is built by the LDA score look-up table which stores all the topical possibilities for each word involved in the vocabulary. Thus, in TDRLM, we replace the original $\mathbf{K}$ during Scaled Dot-Product in the Topical Multihead Attention by our topical score scaled Key. This can effectively avoid topics to confuse ...

CRedit authorship contribution statement

Xinyu Hu: Conceptualization, Methodology, Software, Validation, Writing – original draft. **Weihan Ou:** Software, Writing – review & editing. **Sudipta Acharya:** Writing – review & editing, Supervision. **Steven H.H. Ding:** Conceptualization, Software, Writing – review & editing, Supervision. **Ryan D’Gama:** Software, Data curation. **Hanbo Yu:** Resources. ...

Declaration of Competing Interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Steven H. H. Ding reports financial support was provided by Queens University. ...

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